Solution Requirements

Date	05 July 2026
Team ID	SWTID-2026-4324
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	**Requirement Category**	**Requirement Description**	**Priority (High/Medium/Low)**
1	Authentication	*Users (Farmers/Admin) can securely log in using a username, password, or mobile number with OTP verification.*	High
2	Authorization levels	Admin can manage all users and data. Farmers can access only their own crop, soil, and recommendation information.	High

3	External interfaces	*The system integrates with weather APIs, soil databases, and cloud storage to fetch real-time weather data and support crop recommendations*	High
4	Transactions processing	The system securely processes user inputs such as soil details, crop selection, and weather data to generate optimized crop and fertilizer recommendations.	High
5	Reporting	The system generates reports and dashboards showing crop recommendations, soil analysis, weather forecasts, fertilizer suggestions, and expected yield. Reports can be viewed or downloaded by users.	Medium
6	Business rules, etc.	The system recommends crops and fertilizers based on soil type, weather conditions, season, and available farm resources. Only authenticated users can update or access their farm data.	High
7	Compliance to laws or regulations	*The system follows data privacy and security guidelines. User information is stored securely and used only for agricultural recommendation purposes, complying with applicable data protection regulations.*	*Medium*

8	*Other*	The system provides multilingual support, notifications for weather changes and farming activities, and allows users to download reports for future reference.	Medium

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	The system should load pages within 2–3 seconds and generate crop recommendations within 5 seconds.	High
2	Scalability	The system should support 1,000+ users and handle increasing agricultural data without affecting performance.	Medium
3	Security & Data Privacy	User information and farm data should be securely stored using password encryption, secure login, and role-based access control.	High
4	Reliability & Availability	The system should be available 24×7 with at least 99% uptime and recover quickly from failures.	High

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**

5	Usability & Accessibility	The system should provide a simple, user-friendly interface that is easy for farmers to use on desktop and mobile devices.	Easy navigation, responsive design, and users can complete common tasks without assistance.
6	*Other*	The system should be easy to maintain, update, and deploy on different platforms. It should support future feature enhancements with minimal changes.	Compatible with Windows, Android, and modern web browsers; updates can be deployed without affecting existing functionality.